

GRAND ISLE, LA, USA

WMO: 994977

Lat: 29.2647N

Lon: 89.9576W

Elev: 33

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.0	40.1	18.6	14.0	38.0	24.1	18.3	42.6	32.3	46.4	28.9	46.3	18.5	10	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	89.8	78.4	88.5	78.1	87.4	77.9	80.2	85.4	79.6	84.7	79.1	84.4	6.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	149.4	83.3	78.1	146.2	83.3	77.5	142.9	83.2	43.6	85.0	43.0	84.8	42.5	84.6	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.3	24.5	20.8		DB	32.4	92.8	3.6	3.3	29.8	95.2	27.6	97.2	25.6	99.0	22.9	101.5
				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.1	55.1	58.1	64.0	70.4	76.9	82.1	83.4	83.5	81.5	74.4	65.0	58.7	(6)
	DBStd	11.67	8.27	7.86	6.94	5.07	3.84	2.56	2.41	2.65	3.15	4.95	7.09	8.16	(7)
	HDD50	103	48	22	6	0	0	0	0	0	0	0	3	24	(8)
	HDD65	954	315	209	99	17	1	0	0	0	0	4	87	222	(9)
	CDD50	7819	205	250	439	611	835	962	1034	1040	944	755	451	293	(10)
	CDD65	3193	6	17	67	177	370	512	569	575	494	294	86	26	(11)
	CDH74	30719	0	10	43	434	2765	5951	7105	7212	5373	1686	139	1	(12)
	CDH80	9631	0	0	0	4	270	1926	2783	2902	1592	153	1	0	(13)
(14) Wind	WSAvg	10.0	11.5	11.0	10.6	11.3	9.9	8.3	7.6	7.6	9.0	10.5	11.3	10.9	(14)
(15) Precipitation	PrecAvg	60.6	5.6	3.7	4.6	4.1	3.5	5.7	7.3	6.5	6.5	4.4	3.7	4.2	(15)
	PrecMax	90.1	18.5	10.2	12.8	21.1	7.8	15.0	14.1	12.0	15.6	12.8	14.3	7.0	(16)
	PrecMin	41.3	0.0	0.4	0.4	0.0	0.4	1.5	2.9	3.1	2.6	0.2	0.0	1.4	(17)
	PrecStd	13.6	5.0	2.6	2.8	4.4	2.3	3.3	2.8	2.6	3.3	3.5	3.5	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.5	75.7	77.3	80.5	86.2	89.7	91.3	92.8	90.4	85.4	79.6	73.9	(19)
		MCWB	68.7	69.5	70.8	73.0	73.8	77.6	77.4	82.2	78.0	78.4	76.4	71.4	(20)
	2%	DB	69.2	72.1	75.3	79.0	83.8	88.3	89.2	90.3	87.8	83.1	77.3	72.1	(21)
		MCWB	67.0	68.5	70.7	73.3	75.8	77.0	78.1	79.1	77.0	77.1	73.6	69.7	(22)
	5%	DB	67.6	69.8	73.7	77.7	82.4	87.1	88.0	88.7	86.6	81.7	75.7	70.7	(23)
		MCWB	65.8	67.3	69.6	72.7	75.6	76.6	78.1	78.5	76.9	76.8	72.7	68.5	(24)
	10%	DB	66.0	67.9	72.3	76.7	81.6	86.1	87.0	87.4	85.6	80.4	74.2	69.1	(25)
		MCWB	64.3	65.8	68.7	71.8	75.3	76.8	78.1	78.1	76.8	76.0	71.0	67.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.0	69.1	73.2	75.4	78.9	79.5	80.6	81.8	80.5	79.3	78.0	71.0	(27)
		MCDB	69.7	70.2	74.1	77.3	81.5	83.8	85.3	89.0	83.8	81.3	79.7	71.8	(28)
	2%	WB	67.3	67.9	71.5	74.5	77.7	78.9	80.1	80.3	79.5	78.3	75.7	69.4	(29)
		MCDB	67.9	69.0	72.6	76.7	80.7	83.5	85.2	86.3	83.9	80.7	77.0	70.2	(30)
	5%	WB	65.9	66.4	70.4	73.7	77.1	78.3	79.5	79.6	78.5	77.3	74.2	68.3	(31)
		MCDB	66.8	67.5	71.6	76.1	80.3	83.2	85.0	85.3	83.4	80.1	75.9	69.3	(32)
	10%	WB	64.2	64.9	68.9	72.7	76.4	77.8	79.0	79.1	77.9	76.3	72.5	66.5	(33)
		MCDB	65.2	65.9	70.2	75.4	80.0	83.0	84.7	84.9	83.1	79.5	74.7	67.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	8.8	8.0	6.7	5.9	6.2	6.7	7.0	6.4	7.1	8.4	8.7	(35)
		MCDBR	9.4	8.7	7.3	6.1	6.2	7.0	7.1	8.2	7.4	7.8	7.8	8.2	(36)
	5% WB	MCWBR	8.7	7.1	6.4	4.8	3.5	4.4	4.2	4.5	4.0	3.4	7.2	8.5	(37)
		MCWBR	9.4	9.4	6.8	5.3	5.1	6.2	6.7	6.6	6.1	5.5	7.9	8.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.348	0.363	0.377	0.420	0.444	0.463	0.492	0.492	0.444	0.390	0.364	0.353	(40)
		taud	2.530	2.505	2.449	2.324	2.298	2.291	2.237	2.234	2.333	2.478	2.493	2.528	(41)
	Ebn at Noon		282	287	289	278	271	264	256	255	265	274	274	275	(42)
		Edh at Noon	28	31	35	41	42	43	45	44	39	32	29	27	(43)
(44) All-Sky Solar Radiation	RadAvg		975	1170	1575	1884	2079	1971	1815	1702	1631	1462	1116	877	(44)
	RadStd		79	143	138	170	100	145	126	139	126	138	103	74	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (17 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+286	+191	

Nomenclature: See separate page